

Maritime Search and Rescue Based on Group Mobile Computing for Unmanned Aerial Vehicles and Unmanned Surface Vehicles

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I. INTRODUCTION

Abstract—Accidents often occur at sea, so effective maritime search and rescue is essential. In the current process of sea search and rescue, the operation efficiency of large search and rescue equipment is low and it cannot provide stable communication link. In this article, unmanned aerial vehicles (UAVs) and unmanned surface vehicles (USVs) are used to form a cognitive mobile computing network for co-operative search and rescue, and reinforcement learning (RL) is used to plan search path and improve communication throughput. Based on the scene of marine search and rescue, the grid method is used to model the search and rescue area. Meanwhile, an intragroup communication architecture based on UAVs and USVs is designed to assist intragroup communication by recognizing the link channel state between UAVs. Search and rescue path planning is carried out through the strategy iteration of Markov decision process (MDP). Furthermore, distributed RL is used to recognize the channel state and perform mobile computing, so as to optimize the data throughput in the communication group. The simulation results show that we have successfully completed the path planning task. Compared with conventional methods, RL based on different reward functions has better throughput performance under the same number of UAVs auxiliary communications.

Index Terms—Data throughput, distributed reinforcement learning, markov decision process, path planning.

MARINE transportation is the main mode of international cargo transportation, and more than 80% of the international trade cargo volume is completed by marine transportation every year. However, the marine environment is complex and the weather is changeable. In the course of maritime transportation, marine accidents caused by natural disasters such as gale, huge waves, storms, thunder and lightning, tsunami, and so on often occur, which threatens the world economic development.

At present, almost all the maritime search and rescue activities need human participation, and the search and rescue efficiency is low [1]. As marine communication lacks land infrastructure such as base station, the marine search, and rescue communication system takes satellite as the main communication carrier. Therefore, the image and video information generated in the search and rescue process cannot be transmitted back through the system; so the quality of search and rescue cannot be guaranteed. With the development of intelligent equipments, we propose a novel search and rescue architecture based on unmanned aerial vehicles (UAVs) and unmanned surface vehicles (USVs) and use their mobile computing capabilities to realize the autonomous path planning of equipments in the search and rescue process [2]. Meanwhile, in order to transmit the search and rescue information successfully, we use UAVs and USVs to build a temporary maritime communication network, and use their cognitive communication capabilities to design the intragroup communication forwarding strategy, which improves the intragroup communication throughput [3]. This measure enables UAVs and USVs with limited energy to transmit more data, thus improving the energy efficiency of the system [4].

In this article, we give full play to the mobile computing capabilities of UAVs and USVs to collaborate and build the system model based on reinforcement learning (RL) algorithm [5]. In this system, we use the co-operation of UAVs and USVs to complete the regional exploration, and use multiagent RL algorithm to analyze the channel state between UAVs and USVs and then improve the data throughput in communication group by means of auxiliary cognitive communication. Autonomous path planning is one of the embodiment of UAVs and USVs intelligence. Path planning techniques can be divided into two categories according to information sources: one is global path planning technology based on map information, and the other

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is local path planning technology based on sensor information. In the research of path planning, it also includes a series of technologies such as navigation and map analysis [6]. Based on the characteristics of lack of information on sea maps, we use the multiagent RL method to complete the exploration between task areas through the distributed mobile computing between UAVs and USVs.

In this article, we use RL to study intragroup throughput in mobile ad hoc networks [7]. UAVs and USVs continuously adapt to channel state through cognitive communication of channel information such as signal-to-noise ratio (SNR) in communication group. Through cooperative communication and group mobile computing between UAVs, the data throughput in communication group is improved, and the energy efficiency of the system is optimized. In addition, the Markov decision process (MDP) has been applied to the UAVs and USVs ocean exploration task in this article. Through grid modeling of the exploration area, UAVs and USVs are divided into several groups to carry out the path search task. After that, the ocean exploration task is completed through case analysis and summary of brain-inspired control center.

Aiming at the problem of exploratory communication in unknown marine environment, RL method has attracted wide attention because of its self-adaptability and no need for a large number of prior knowledge. Especially in the field of ocean exploration, which lacks environmental information, it stands out in a series of machine learning methods. In this article, RL is used to model the ocean environment exploration, and RL algorithm is used to optimize the data throughput of the system. These two measures improve the energy efficiency of the system. A novel distributed mobile computing ocean exploration model and ad hoc group network architecture are established. The main contributions of this article are as follows:

- 1) A seaborne region exploration architecture based on UAVs and USVs joint grouping is proposed. The architecture is divided into air and surface layers, and equipped with brain-inspired control centers. The concepts of centralized control and distributed mobile computing are adopted to decouple the control and forwarding functions of the network. Search and rescue task is completed by UAVs and USVs grouping path planning. A novel intragroup communication model of ad hoc network is designed to improve intragroup throughput through UAV interconnection.
- 2) MDP is used to model the path planning problem, and the strategy iteration method is used to solve the problem. In addition, we use distributed RL algorithm to optimize the throughput of ad hoc group. Through these measures, the system can complete more exploration and forwarding tasks under the same energy condition, thus improving the energy efficiency of the system.
- 3) We design the path planning reward function based on MDP and the throughput reward function based on distributed RL algorithm. The validity of MDP method in path planning task is verified by simulation. Meanwhile, distributed RL method improves system throughput significantly.

The rest of this article is organized as follows. System model is given in Section III. Algorithms is proposed in Section IV. In Section V, the simulation results show that the MDP has successfully completed the task of marine area exploration and has strong environmental adaptability. At the same time, using the reward function designed by us, the RL algorithm successfully improves the system throughput and optimizes the energy efficiency in the communication group. Finally, Section VI concludes this article.

II. RELATED WORK

In Section II, we discuss some related works. The development of intelligent equipment such as UAVs and USVs provides us with new ideas for maritime search and rescue mission. By reasonably utilizing the distributed mobile computing capability of intelligent devices, we can design UAVs and USVs collaboration system models based on task types. Sato and Tachibana [8] proposed a selective recognition method for heterogeneous multiagent tasks, which reduced the difficulty of complex tasks. Wang and Feng [9] solved the problems of co-operative simulation system of autonomous underwater vehicle by using multiagent co-operation module, and proposed a co-operative decision rule among multiagents. Mao *et al.* [10] proposed a multiagent algorithm for wireless sensor networks and the Internet of Things, which improved energy consumption. The environmental adaptability of multiagent models has been fully validated in these studies.

In the face of some unexplored and disastrous environments, we often make full use of the autonomous cognitive characteristics of UAVs and USVs. Combining UAVs and USVs system with RL algorithm can often achieve unexpected application results. Prasad and Ramasangu [11] constructed the communication topology using multiagent tree structure and formed the tree by RL method, which solved the multiagent induction problem when the target state was unknown. Sause *et al.* proposed a framework for coordinating multiple agents in a single virtual environment. The framework was implemented by Q-learning and Sarsa time-difference RL algorithm, and proved that the Sarsa method has better practicability [12]. Multiagent system also promotes the research in the field of unmanned driving. Liu *et al.* have studied the experience replay method based on fully distributed decision-making. The method is improved by using deep RL technology and its advantages are verified [13]. They also studied UAV control methods based on deep RL [14]. All these studies have proved that RL method is compatible with multiagent system.

Faced with the lack of offshore energy supply means, in the process of maritime search and rescue, besides considering system performance, such as throughput, delay, and so on, the energy efficiency management of UAVs and USVs is also a research emphasis. Wei *et al.* [15] make use of the characteristics of ocean channel and the mobility of communication users to improve energy efficiency and significantly reduce communication power consumption. Mochaourab *et al.* [16] studied the distributed coordination algorithm to improve channel awareness and access performance. Lu *et al.* [17] studied the joint

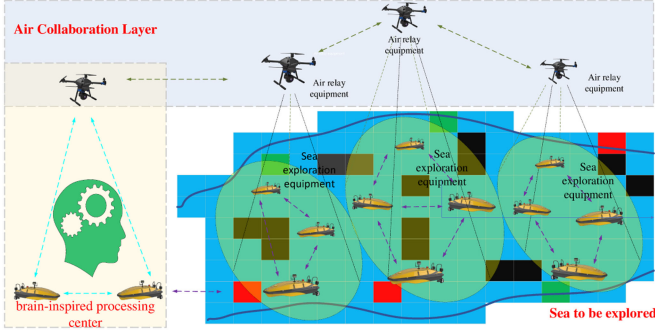


Fig. 1. Search and rescue exploration model based on UAVs and USVs collaboration.

optimization of subcarrier and power allocation in 5G wireless networks to maximize the energy efficiency of multiuser systems. Using multiagent RL method to analyze the energy efficiency problem in the process of multiagent communication becomes the basis of the research on energy efficiency of multiagent at sea. N. *et al.* [18] studied the distributed behavior of mobile users and proposed a multihop data transmission routing method for heterogeneous networks based on RL, which realized high-capacity path transmission.

III. SYSTEM MODEL

Based on the background of maritime search and rescue, this article designs the system model based on the mobile computing of USVs and UAVs, in which the maritime search and rescue mission is divided into two parts. The first part is the search path planning of marine search and rescue equipment. To this end, we established a joint search and rescue model based on UAVs and USVs, and used mobile computing to complete the exploration mission in the accident area. In addition, aiming at the communication tasks in the rescue process, based on cognitive communication abilities of USVs and UAVs we design an intragroup communication model to improve the intragroup data throughput in mobile ad hoc networks [19].

A. Maritime Search and Rescue Model Based on Group Mobile Computing

For the search and rescue task, we use the method of grid division to represent the map in the exploration of path planning, which avoids the complicated calculation in dealing with the obstacle boundary of obstacles [20]. The grid in the figure represents the sea area being explored, and we complete this exploration process through UAVs and USVs group collaboration. Among them, the USV team is responsible for sea surface exploration, and the aerial UAV is responsible for communication control and dispatching work [21]. The system model is shown in Fig. 1. In the figure, the black grid represents the obstacles in the sea area, the red grid indicates the starting point of the area exploration, and the green indicates the end point of the area exploration. Each group has its own starting point and ending point. It can be seen from the figure that the sea surface USVs are responsible for the area exploration and coverage work, and

they can use the MDP to complete the environment perception and exploration tasks, and achieve this goal through intergroup communication coordination.

The information processing in the whole exploration process is completed by the brain-inspired control center, which stores part of the known environmental information, and can update its case base continuously through exploration actions to achieve a conditional reflection action response. After the end of a reinforcement study, the case base analyzes whether it needs to be updated according to the strategic effect, compares the case with higher similarity in the action selection stage of the RL, and judges whether to reuse the case to guide the action selection. This mechanism is essential for us to search and rescue, so that we can enrich the existing knowledge base based on the information we have explored, and improve intelligence and search efficiency in the subsequent exploration process, while avoiding unnecessary search and improving energy efficiency.

B. Cooperative Communication Model in UAVs and USVs Groups

At present, due to the lack of basic resources for maritime communication and communication infrastructure such as land-based base stations, UAVs and USVs system is difficult to return to the collected images and videos when performing maritime search and rescue and exploration tasks [22]. In the communication group designed in this article, because the USVs and the UAVs have different characteristics, their communication functions are different. Due to its large size, the USV can store more energy and load larger transmission power [23]. Therefore, the USVs plays the role of the central base station in the communication group and is responsible for information forwarding and processing. The UAVs can participate in the interconnection between the UAVs and the communication with the USV in the communication group. Therefore, the intragroup communication UAVs are divided into two groups: A2S (UAV to USV) communication and A2A (UAV to UAV) communication. The intragroup collaborative communication model is shown in Fig. 2. In the uplink for packet backhaul, we assume that the group's UAVs share the uplink spectrum resources and reuse the resource blocks in the group. According to Shannon's theorem, the limit, error-free information transmission rate of a channel with limited bandwidth and Gaussian white noise interference is

$$C = B \log_2(1 + S/N) \quad (1)$$

where B is the channel bandwidth, S is the average power of the transmitted signals in the channel, and N is the Gaussian white noise power inside the channel. From the formula we can see that under the condition of limited bandwidth, the channel performance can only be improved by increasing the signal-to-noise ratio (SNR) parameter. Therefore, in a system, the link quality of the communication link in which the resource block is located is usually measured by the SNR. The SNR calculation equation is

$$\text{SNR} = \frac{p_{\text{signal}}}{p_{\text{noise}}} = \left(\frac{A_{\text{signal}}}{A_{\text{noise}}} \right)^2 \quad (2)$$

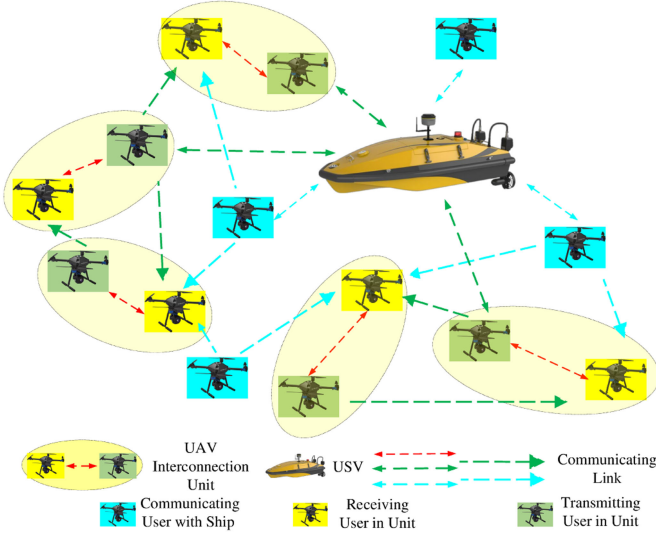


Fig. 2. UAVs and USVs communication model for auxiliary communication through A2A.

where p_{signal} is the signal power, p_{noise} is the noise power, and A_{signal} is the corresponding signal amplitude, A_{noise} is the signal amplitude. Since the users of A2S use orthogonal resources, there is no channel interference noise, and there is mutual interference between the users participating in the A2S and the users of the A2A. In the system we designed, each resource block is allowed to link a A2S user and several A2A users. We deduce that SNR when the x th UAV mutual transfer user uses the n th resource block is

$$S_n^{U_x} = \frac{P_n^{U_x} \cdot G_{xx,n}^{U_x}}{\sum_{y \neq x, y \in L_n} P_n^{U_y} \cdot G_{yx,n}^{U_x} + P_n^e \cdot G_n^{ex}} \quad (3)$$

where n indicates that these parameters are all defined based on the resource block n , $P_n^{U_x}$ indicating the transmit power of the x th participating A2A user on the resource block, P_n^e is the transmit power of the A2S user. $G_{xx,n}^{U_x}$ is the channel gain on the x th A2A link, and $G_{yx,n}^{U_x}$ is the channel gain of the receiving user on the y th A2A link to the receiving user on the x th A2A link, G_n^{ex} indicates the channel gain of the e th A2S user to the x th A2A user. L_n represents the set of links of the A2A users sharing the n th resource block. Based on the above information, we can define the SNR of the e th A2S user on the n th resource block as

$$S_n^e = \frac{P_n^e \cdot G_{e,n}^0}{\sum_{y \in L_n} P_n^{U_y} \cdot G_{y,n}^{y0}} \quad (4)$$

where, $G_{e,n}^0$ represents the channel gain between the UAVs and the USVs of the A2S user on the n th resource block, and $G_{y,n}^{y0}$ represents the channel gain between the A2A users and the base stations.

IV. PROPOSED ALGORITHM

In this article, distributed multiagent RL technology is used to realize ocean exploration and communication scheduling.

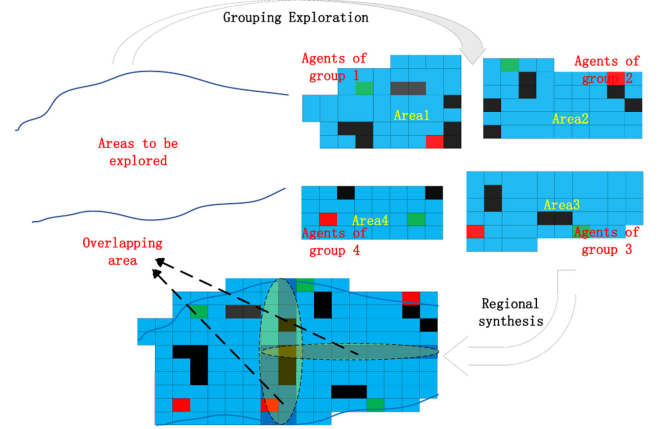


Fig. 3. Ocean search and rescue task is accomplished through distributed exploration of different UAVs and USVs teams and calculation of brain-inspired control center.

In view of the marine environment, we give full play to the mobile computing ability of intelligent devices, carry out cognitive communication within the group, and make full use of the upstream spectrum resources to solve the exploration of unknown marine environment and network optimization problems.

A. UAVs and USVs Path Planing Based on Group Mobile Computing

We divide the UAVs and USVs system into different groups to explore the sea area [24]. First of all, we will randomly divide the area. Therefore, the exploration information retained on each control UAV or USV is different, and it is part of the entire map information. The air control UAV can complete the puzzle of the entire exploration area through communication and exchange the information to the brain-inspired control center for storage processing. The exploration model is shown in the following Fig. 3. The sea area to be explored is modeled as a grid area. The task of the agent is to reach the end point from the starting point, and there are obstacles in the path between the starting point and the end point. The agent can follow the different paths to reach the end point from any starting point in the grid. Different paths correspond to different rewards. Our planning goal is to find the path with the highest reward as the shortest path.

State Space: In the path planning problem, the state space of the agent comes from all the cells. At any moment, the agent will be in a certain state of the whole state space. The state of the agent at time t is represented as s_t , and the whole state space is recorded for S , $s_t \in S$. In order to be able to reach the target state s_n (the end of the path) from the initial state s_0 , the agent must take a series of actions, denoted by $a_0, a_1, \dots, a_{n-1}$.

Action Space: At any t time, the agent can select any action a_t from the action space A to perform cell transfer. The set of actions allowed at time t depends on the state s_t in which the agent is located, including one or several sets of $a \in [\text{east, west, south, north}]$.

Strategy: When the agent takes action, it will enter the next cell. The state s_{t+1} occupied by the agent at time $t+1$ depends

on s_t and a_t , as follows:

$$s_{t+1} = f(s_t, a_t). \quad (5)$$

Reward Function: In the shortest path problem, the agent executes action a_t in state s_t at time t and shifts to new state s_{t+1} . We define the reward function as

$$r_t = \begin{cases} 100, \text{ endpoint cell reward} \\ 0, \text{ non-boundary cell reward} \\ -1, \text{ boundary cell reward} \\ -100, \text{ barrier cell reward} \end{cases}. \quad (6)$$

Then we calculate the value of each state according to the transition probability, strategy and reward iteration equation

$$\begin{aligned} V^\pi(s) &= \sum_a \pi(s, a) \sum_{s'} P_{ss'}^a (R_{ss'}^a + \gamma V^\pi(s')) \\ &\times \left(R_{ss'}^a + \gamma E_\pi \left\{ \sum_{k=0}^{\infty} \gamma^k r_{k+t+2} | s_t = s \right\} \right) \\ &= \sum_a \pi(s, a) \sum_{s'} P_{ss'}^a (R_{ss'}^a + \gamma V^\pi(s')) \end{aligned} \quad (7)$$

then we calculate the state-behavior value function is

$$Q^\pi(s, a) = \sum_{s'} P_{ss'}^a (R_{ss'}^a + \gamma V^\pi(s')) \quad (8)$$

$$Q^*(s, a) = E\{r_{t+1} + \gamma \max_{a'} Q^*(s_{t+1}, a') | s_t = s, a_t = a\} \quad (9)$$

and we can search for maximizing behavior. If multiple behaviors can generate equivalent maximization results, then assign equal probability to each behavior. So the state value we calculated is based on the optimal behavior strategy, and the strategy iteration algorithm is shown in Algorithm IV-A.

B. Energy Efficiency Optimization Algorithm Based on Distributed RL Learning

In a communication group, each A2A user x performs independent calculations and maintains a separate table Q , $Q^x(s, a)$. Since the entire system uses distributed computing, each agent is only responsible for a part of the computing tasks of the entire system. The system single Q table size depends on the number of agents and the status and action space.

State Space: We define the state space of the system according to the noise level existing on the resource block n at time t

$$W_t^n = \begin{cases} 1, S_n^e \geq \gamma^0 \\ 0, \text{ otherwise} \end{cases} \quad (10)$$

where γ^0 is the minimum SNR that guarantees the quality of service of the A2S users. Thus, the state of the A2A user x on the k th resource block can be defined as

$$S_t^{x,n} = \{W_t^n\}. \quad (11)$$

Algorithm 1: Agent Exploration Algorithm Based on Strategy Iteration.

```

1: Initialize:  $V(s) = 0$ , for all  $s \in S$ 
2: Evaluate the strategy.
3: repeat
4:    $\Delta = 0$ 
5:   for each  $s \in S$ :
6:      $v \leftarrow V(s)$ 
7:      $V(s) \leftarrow \sum_a \pi(s, a) \sum_{s'} P_{ss'}^a (R_{ss'}^a + \gamma V^\pi(s'))$ 
8:      $\Delta = \max(\Delta, |v - V(s)|)$ 
9:   until  $\Delta < \theta$ , for all  $s \in S$ , ( $\theta$  is a very small positive number.)
10: Improving the Strategy.
11: repeat
12:   for each  $s \in S$ :
13:      $b \leftarrow \pi(s)$ 
14:      $\pi(s) \leftarrow \max_a \sum_{s'} P_{ss'}^a (R_{ss'}^a + \gamma V^\pi(s'))$ 
15:   if  $b \neq \pi(s)$  then
16:     Return strategy evaluation
17:   end if
18: until  $b \neq \pi(s)$ 

```

In this article, we assume that the A2S user will share its SNR ratio information to the USV and the A2A user who shares a resource block with it.

Action Space: We define the action space as the discrete power set optional by A2A user x

$$A_x = (a_1^n, a_2^n, \dots, a_l^n) \quad (12)$$

where l indicates that the A2A user has l discrete power values.

Reward Function: According to our analysis in the system model, in order to ensure the reliability of data transmission, this article should give priority to ensuring the channel transmission quality of the A2S users. Although the optimization goal in this article is to maximize the system throughput, this goal should be based on the premise of ensuring the channel quality of the A2S users. Therefore, we assume that the A2S user's transmit power is fixed, and the system throughput optimization target can be defined as

$$\begin{aligned} &\max_{p_n} \sum_{n=1}^n \{ \log_2(1 + S_n^e) + \sum_{x \in L_n} \log_2(1 + S_n^{U_x}) \} \\ &\text{s.t. } 0 \leq p_n^{U_x} \leq p_{\max}, \forall n, x \\ &S_n^e \geq \gamma^0 \end{aligned} \quad (13)$$

where P_n represents the transmit power of the A2A user on the n th resource block. According to the system objective, we use the total throughput on the n th resource block as a reward function

$$r_1^{x,n} = \log_2(1 + S_n^e) + \sum_{x \in L_n} \log_2(1 + S_n^{U_x}) \quad (14)$$

the calculation of total throughput requires signaling inter-transmission among multiple agents, which will occupy a certain amount of channel overhead and reduce system energy

Algorithm 2: Multi-agent Distributed Reinforcement Learning Algorithm.

```

1: Initialize:  $Q^x(s, a) = 0$ , for all  $s \in S, a \in A$ 
2: Evaluate  $S_0$ .
3: loop
4:   Generate a random number  $c, c \in (0, 1)$ 
5:   if  $x < \varepsilon$  then
6:     Random selection
7:   else
8:     Select the action  $a, a \in A$ , to maximize Q value
9:   end if
10:  Execute action a, receive the immediate reward  $r^{x,n}$ 
11:  Enter the next state, update Q value table
12: end loop

```

efficiency. We adopt a distributed method to maximize the throughput of each A2A user to achieve our optimization goals and improve the energy efficiency of the system. Therefore, according to the throughput of each connected A2A user, we define the reward function as follows:

$$r_2^{x,n} = \begin{cases} \frac{1}{c} \log_2(1 + S_n^{U_x}), & S_n^e \geq \gamma^0 \\ -1 & \end{cases} \quad (15)$$

Then the system performs distributed learning and iterates the Q value table according to (16) to obtain the optimal return until the end of the loop

$$Q_{t+1}(s_t, a_t) = \begin{cases} \max\{Q_t(s_t, a_t), r_{t+1} + \gamma \max_{a_{t+1}} Q_t(s_{t+1}, a_{t+1})\} \\ Q_t(s_t, a_t), & \text{otherwise} \end{cases} \quad (16)$$

V. SIMULATION RESULTS AND DISCUSSION

Aiming at the problem of multiagent maritime region exploration, we designed collaborative regional exploration strategy, and the strategy value iterative algorithm was used to find the optimal path. In addition, in view of the lack of basic communication resources in the exploration process, we optimize the throughput rate of the system through mobile computing of UAVs and USVs to improve energy efficiency. We simulated the algorithm in the MATLAB R2014a and Python environment, and established simulation regions for different situations to analyze the performance of the algorithm.

A. Regional Collaboration Search and Rescue Simulation

We modeled the area to be explored as a grid, set up obstacles according to the actual situation of the sea area, and set the starting point and the end point. According to the reward function we designed, the reward of the end point is set to 100, and the reward of the obstacle area is set to -100 . As shown in Fig. 4, it is a grid area of 10×10 , with blue as the starting point, green as the ending point, and red as the obstacle area. Fig. 4 shows the different states of the agent during the path exploration process. According to the MDP, the strategy is iteratively updated to

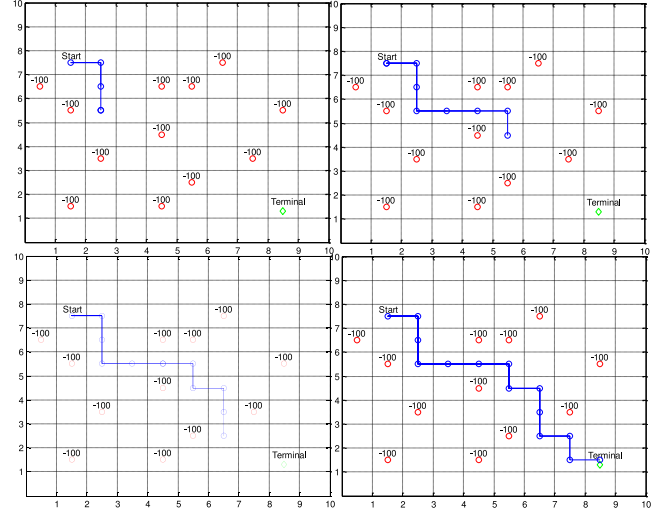


Fig. 4. UAVs and USVs regional exploration and path planning process based on MDP.

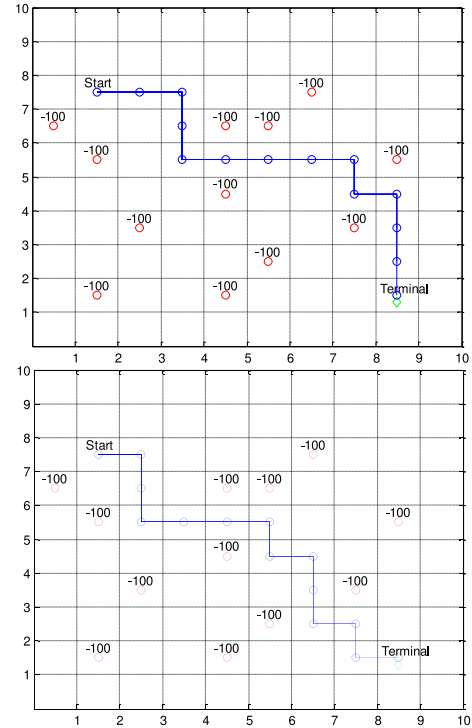


Fig. 5. Same path planning task based on different state transition probabilities.

optimize the best strategy in the exploration process. We set the discount factor γ to 0.95 and the state transition probability P to 0.8. It can be seen from Fig. 4 that the final agent successfully explores the end point and avoids the obstacle area.

Since the maritime agent has limited energy carrying capacity, energy can be saved if the direction is changed less during the regional exploration process. Therefore, In the case of ensuring the path length is consistent, we change the exploration path by changing the state transition probability. In the case shown in Fig. 5, we set the state transition probability P to 0.6, and

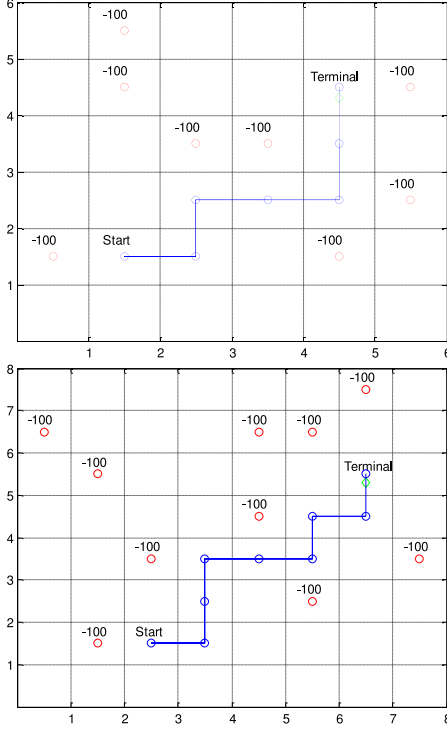


Fig. 6. Examples of path planning tasks based on different grid granularity.

TABLE I
SIMULATION PARAMETERS

Relevant parameters of communication	Value
Constant c in equation 15	40
Distance between A2A teams	0.05 n mile
The smallest A2S users' S/N , γ^0	7dB
Point Source Antenna Gain of USV	18dBi
Point Source Antenna Gain of UAV	5dBi
Maximum transmit power for A2A users	25dB
Noise interference power	-117dB
Single resource block bandwidth	180kHz
Path Loss Model between UAVs	$15+38 \log_{10}(d(km))$
Path Loss Model between UAV and USV	$28+40 \log_{10}(d(km))$

obtained a lesser path with less change in direction, as shown in the top of Fig. 5. Besides, in the process of exploring irregular regions, we need to divide different grid granularity according to the actual conditions. Fig. 6 shows the region exploration method with grid granularity of 6×6 and 8×8 , which proves the validity and applicability of the MDP process.

B. Simulation of System Data Throughput

In this article, we simulate according to the single-group scenario shown in Fig. 2, and the various parameter settings in the communication system are shown in Table I. According to the communication scenario shown in Fig. 2, combined with the communication capabilities of USVs and UAVs, this article sets the coverage area radius to 0.1 nautical miles and the number of agents to 50. We set the number of resource blocks in the communication system based on the number of A2S users. In the process of applying the RL algorithm, we set the discount

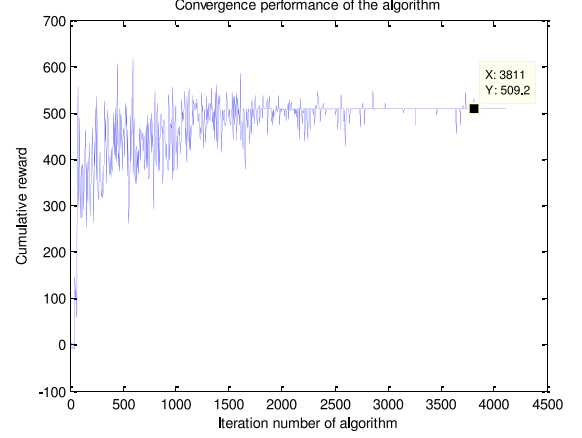


Fig. 7. Convergence performance of RL algorithm based on reward $r_1^{x,n}$.

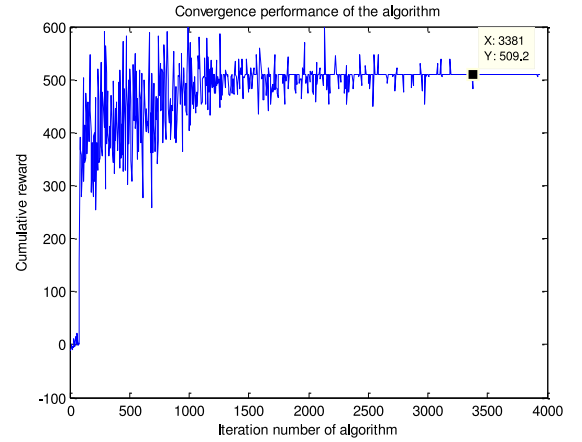


Fig. 8. Convergence performance of RL algorithm based on reward $r_2^{x,n}$.

factor γ to 0.9, the learning rate α to 0.8, and the epsilon value for the greedy algorithm is 0.9.

First, we performed the convergence performance simulation of the distributed RL algorithm, as shown in Figs. 7 and 8. In the two figures we adopt different reward functions $r_1^{x,n}$ and $r_2^{x,n}$, where Fig. 7 uses $r_1^{x,n}$ and Fig. 8 uses $r_2^{x,n}$. It can be clearly seen from the legend that when it converge to the same reward, the reward function based on A2A users can converge faster. Since the A2A user only needs to maintain a smaller Q -value table, there is a faster convergence speed.

Then we simulated the throughput of A2A users under different reward functions, using the method based on open-loop power control as a comparison. From Fig. 9, we can see that the throughput of A2A users increases with the number of users, and the reward function can achieve higher throughput in the case of more A2A users. Fig. 10 reflects the system throughput of the whole communication group. It can be observed that the reward function $r_1^{x,n}$ works better across the system because it considers parameters of the whole system, but it converges slowly. Therefore, according to the number of A2A users, we can flexibly select the reward function to achieve the balance of

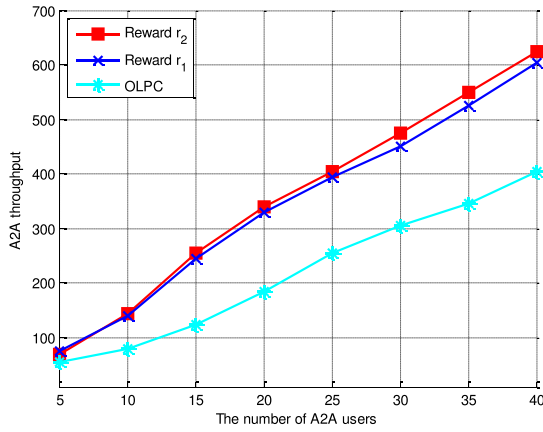


Fig. 9. A2A throughput based on different rewards.

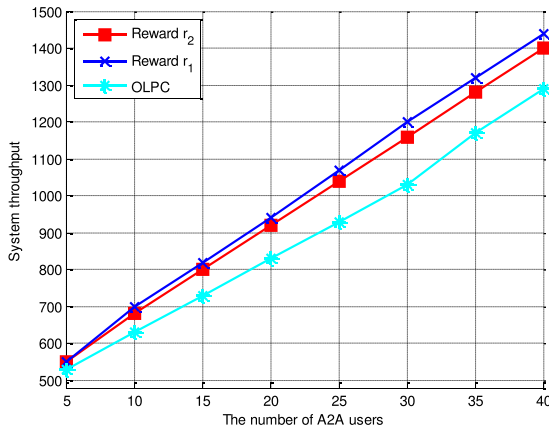


Fig. 10. System throughput based on different rewards.

convergence performance and throughput in actual use, so as to improve the energy efficiency of the system.

VI. CONCLUSION

A solution for search and rescue problems at sea was proposed in this article, including exploration path planning and ad hoc group networking methods. MDP was used to explore unknown sea areas, then we distributed the exploration tasks of unknown sea areas to different groups of UAVs and USVs, collaborated to complete path planning through mobile computing of UAVs and recorded exploration models through brain-inspired control center to enrich the knowledge base and accelerated the subsequent exploration process. In addition, in the process of ad hoc group network communication, the RL algorithm was fully utilized. By designing different reward functions, we analyzed the throughput change in the cognitive communication process within the group, found the appropriate intra-group communication reward function, and improved the data throughput of the system. These methods took advantage of the social collaborative nature of the maritime UAVs and USVs search and rescue system. In future research, more attention will be paid to energy efficiency indicators of the system to improve system energy efficiency.

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